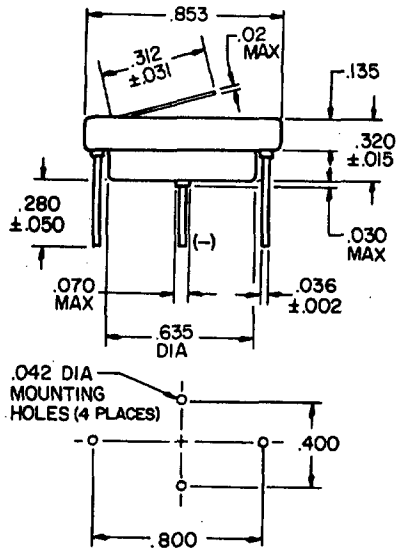
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM ANODE AND CATHODE, STYLES CRL01 AND CRL02

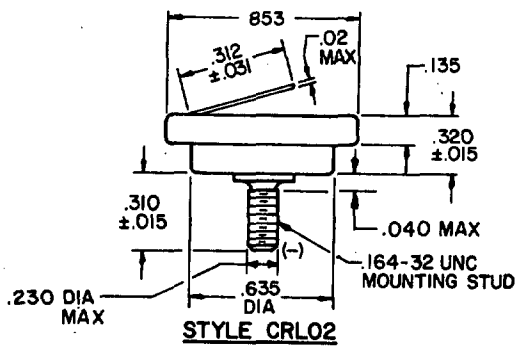
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-83500.



STYLE CRL01

INCHES	MM	INCHES	MM
.002	.05	.070	1.78
.015	.38	.135	3.43
.02	.50	.280	7.11
.030	.76	.312	7.92
.031	.79	.320	8.13
.036	.91	.400	10.16
.042	1.07	.635	16.13
.050	1.27	.800	20.32
		.853	21.67



STYLE CRL02

INCHES	MM
.015	.38
.02	.50
.031	.79
.040	1.02
.135	3.43
.164	4.17
.230	5.84
.310	7.87
.312	7.92
.320	8.13
.635	16.13
.853	21.67

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm).

FIGURE 1. Styles CRL01 and CRL02 capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

DC rated voltage	Cap. (nom.)	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	Impedance (max) -55°C	Capacitance change			Part number M83500/01-	
			+25°C	+85°C +125°C			-55°C	+85°C	+125°C	Style CRL01	Style CRL02
Volts	µF	± percent	µA	µA	percent	ohms	percent	percent	percent		
6	1200	20	8	24	144	20	-90	+30	+35	0001	1001
8	1000	20	8	24	120	20	-90	+27	+30	0002	1002
10	820	20	8	24	100	20	-85	+22	+25	0003	1003
15	680	10	8	24	85	20	-85	+17.5	+20	0004	1004
15	680	20	8	24	85	20	-85	+17.5	+20	0005	1005
20	560	10	5	24	75	25	-85	+17.5	+20	0006	1006
20	560	20	5	24	75	25	-85	+17.5	+20	0007	1007
20	470	10	5	24	72	25	-85	+12.5	+15	0008	1008
20	470	20	5	24	72	25	-85	+12.5	+15	0009	1009
20	390	10	5	24	60	30	-80	+12.5	+15	0010	1010
20	390	20	5	24	60	30	-80	+12.5	+15	0011	1011
30	330	10	5	24	48	30	-75	+12.5	+15	0012	1012
30	330	20	5	24	48	30	-75	+12.5	+15	0013	1013
30	270	10	5	24	40	30	-70	+12.5	+15	0014	1014
30	270	20	5	24	40	30	-70	+12.5	+15	0015	1015
50	220	10	3	24	33.6	35	-70	+12.5	+15	0016	1016
50	220	20	3	24	33.6	35	-70	+12.5	+15	0017	1017
50	180	10	3	24	27.6	35	-65	+12.5	+15	0018	1018
50	180	20	3	24	27.6	35	-65	+12.5	+15	0019	1019
50	150	10	3	24	22.8	40	-65	+10	+12.5	0020	1020
50	150	20	3	24	22.8	40	-65	+10	+12.5	0021	1021
75	120	10	3	24	25.7	45	-55	+10	+12.5	0022	1022
75	120	20	3	24	25.7	45	-55	+10	+12.5	0023	1023
75	100	10	3	24	23	50	-40	+10	+12.5	0024	1024
75	100	20	3	24	23	50	-40	+10	+12.5	0025	1025
75	82	10	3	18	20.7	55	-30	+10	+12.5	0026	1026
75	82	20	3	18	20.7	55	-30	+10	+12.5	0027	1027
75	68	10	3	15	19	60	-30	+8	+10	0028	1028
75	68	20	3	15	19	60	-30	+8	+10	0029	1029
100	56	10	2	20	17.5	60	-25	+8	+10	0030	1030
100	56	20	2	20	17.5	60	-25	+8	+10	0031	1031
125	47	10	5	50	15	60	-20	+8	+10	0032	1032
125	47	20	5	50	15	60	-20	+8	+10	0033	1033

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tantalum anode and cathode with outer jacket of nickel.
Terminals - See figure 1.
Rated temperature - -55°C to +85°C, voltage derated to +125°C.
Rated voltage - See table I.

DC leakage (DCL) (max) (at +25°C): See table I.

Capacitance (Cap.) (nom): See table I.
Cap. tolerance: ±10% or ±20% as shown in table I.

Dissipation factor (DF) (max): See table I.

Solderability: In accordance with MIL-C-83500.
Terminal tabs for styles CRL01 and CRL02.
Radial pin terminals for style CRL01.

Stability at low and high temperatures:
Step 1 (+25°C):
DCL (max) - See table I.
Cap. - Within tolerance of table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55°C):
Impedance - See table I.
ΔCap. - See table I.
Step 3 (+25°C):
DCL (max) - See table I.
ΔCap. - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85°C):
DCL (max) - See table I.
ΔCap. - See table I.
DF (max) - Shall not exceed initial requirement.
Step 5 (+125°C):
DCL (max) - See table I.
ΔCap. - See table I.
DF (max) - Shall not exceed initial requirement.
Step 6 (+25°C):
DCL (max) - See table I.
ΔCap. - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Barometric pressure (reduced):
In accordance with MIL-C-83500.
DC rated voltage applied.

Reverse voltage: In accordance with MIL-C-83500.

Life at 85°C:
2,000-hr (qualification):
DCL (max) - See table I (85°C and 25°C).
ΔCap. - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial requirement.
240-hr (group B):
DCL (max) - See table I (85°C and 25°C).
ΔCap. - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial measurement.
6,000- and 10,000-hr (group C):
DCL (max) at 85°C - Shall not exceed 125% of 25°C value of table I.
DCL (max) - See table I (25°C).
ΔCap. - Within +10, -20% of initial measurement.
DF (max) - 150% of initial requirement.

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Life at 125°C:
DCL (max) - See table I (125°C and 25°C).
ΔCap. - Within +10, -20% of initial measurement.
DF (max) - 150% of initial requirement.

Vacuum life at 125°C:
DCL - See table I (125°C and 25°C).
ΔCap. - Within ±20% of initial measurement.
DF - See table I.
Weight loss - 4.1 milligrams max.

AC ripple life at 125°C:
DCL - See table I (125°C and 25°C).
ΔCap. - Within ±15% of initial measurement.
DF - See table I.

Marking: In accordance with MIL-C-83500.
Part number: M83500/01-(dash number from table I).
Polarity: Plus (+) symbol marked on positive terminal.

Custodian:
Air Force - 85

Review activity:
Air Force - 80

Preparing activity:
Air Force - 85

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